

Gaussian mixture models and geospatial data [MCQ] (Version : 0)

TEST

● **Correct Answer**

🕒 Answered in 3.02 Minutes

Question 1/20

You are tasked with clustering customer data using a Gaussian mixture model (GMM). Which type of clustering does GMM perform?

☐ Hierarchical clustering

☐ Hard clustering

☐ Linear clustering

☒ Soft clustering

Explanation:

- Soft clustering: Correct. GMM performs soft clustering, assigning probabilities of cluster membership to each data point.
- Hard clustering assigns each data point to exactly one cluster.
- Linear clustering is not a recognised term in clustering.
- Hierarchical clustering builds a hierarchy of clusters.

Question 2/20

When using GMM for market segmentation, you notice some customers belong partially to multiple segments. What does this indicate?

☐ There is only one segment

☐ The algorithm is malfunctioning

☐ Customers are incorrectly segmented

☒ GMM assigns probabilities to cluster memberships

Explanation:

- GMM assigns probabilities to cluster memberships: Correct. GMM assigns probabilities, indicating that customers can belong to multiple segments to varying degrees.
- The algorithm is malfunctioning: This is not an error; it's a feature of GMM.
- Customers are incorrectly segmented: Partial membership is correct in GMM.
- There is only one segment: The presence of multiple segments is expected.

Question 3/20

In a GMM, what mathematical concept is used to represent the distribution of each cluster?

☐ Poisson distribution

☒ Gaussian distribution

☐ Uniform distribution

☐ Exponential distribution

Explanation:

- Gaussian distribution: Correct. GMM uses Gaussian (normal) distributions for clusters.
- Exponential distribution is not used in GMM.
- Uniform distribution is not used in GMM.
- Poisson distribution is not used in GMM.

Question 4/20

A data scientist wants to apply GMM to a dataset with non-spherical clusters. Which of the following statements is true?

☐ GMM requires the data to be linear

☒ GMM can handle non-spherical clusters

☐ GMM can only handle spherical clusters

☐ GMM is not suitable for clustering

Explanation:

- GMM can handle non-spherical clusters: Correct. GMM is suitable for non-spherical clusters, unlike k-means.
- GMM can handle non-spherical shapes.
- GMM does not require linear data.
- GMM is suitable for clustering

Question 5/20

When starting the GMM algorithm, how are the initial Gaussian parameters chosen?

☐ By manual selection

☐ By using k-means clustering results

☒ Randomly

☐ By sorting the data

Explanation:

- Randomly: Correct. Initial Gaussian parameters in GMM are chosen randomly.

- K-means results can be used but are not required.
- Manual selection is not the standard approach.
- Sorting the data is not related to initial parameter selection.

Question 6/20

What purpose does the Expectation-Maximisation (EM) algorithm serve in the Gaussian Mixture Model (GMM)?

☐

Assigning data points to clusters based on their likelihood.

☐

Calculating the probability density function of the data.



Updating the Gaussian parameters to best fit the data.

☐

Initialising the parameters of the Gaussian components.

Explanation:

Correct Answer:

Updating the Gaussian parameters to best fit the data.

Incorrect answers:

Initialisation: Initialisation takes place outside of the EM algorithm.

Assigning data points to clusters: This takes place in the EM algorithm, but it's not the sole purpose of the algorithm.

Calculating the probability density function of the data.
Incorrect, as the EM algorithm does not directly calculate the probability density function of the data, but rather estimates the parameters of the Gaussian components that best describe the data distribution.

Question 7/20

You are using GMM to cluster a high-dimensional dataset. How is the covariance matrix represented for each cluster?

☐ As a vector

☐ As a diagonal matrix

☒ As a full matrix

☐ As a scalar

Explanation:

- As a full matrix: Correct. In high dimensions, the covariance matrix for each cluster is a full matrix.

- A scalar does not represent multidimensional covariance.

- A vector is insufficient for covariance in multiple dimensions.

- A diagonal matrix is a simplification, not always used in GMM.

Question 8/20

In a GMM application, you observe that one Gaussian distribution's likelihood dominates due to more data points. How can this be adjusted?

☐ Ignore the dominating distribution

☐ Reduce the number of clusters

☐ Increase the number of clusters

☒ Use weight values to balance the distributions

Explanation:

- Use weight values to balance the distributions:
Correct. Weight values adjust for different numbers of observations in each cluster.
- Reducing clusters does not solve the imbalance.
- Increasing clusters does not address the issue.
- Ignoring the dominating distribution is not a solution.

Question 9/20

While running the Expectation-Maximization algorithm in GMM, what is the purpose of the Expectation step?



To calculate the probabilities of data points belonging to each cluster



To update the Gaussian parameters



To terminate the algorithm if convergence is reached



To assign data points to the nearest cluster

Explanation:

- To calculate the probabilities of data points belonging to each cluster: Correct. The Expectation step calculates the probabilities of cluster membership for each data point.
- Assignment to the nearest cluster is not part of the Expectation step.
- Updating Gaussian parameters occurs in the Maximization step.
- Termination based on convergence is not the purpose of the Expectation step.

Question 10/20

You have applied GMM to a dataset and obtained the following probabilities for three clusters for a particular data point: 0.5, 0.3, and 0.2. To which cluster should this data point be assigned if a hard assignment is required?

☐ Cluster 3

☐ None, since soft assignment should be used

☒ Cluster 1

☐ Cluster 2

Explanation:

- Cluster 1: Correct. The data point is assigned to the cluster with the highest probability, which is Cluster 1.
- Cluster 2 has a lower probability.
- Cluster 3 has the lowest probability.

- While GMM typically uses soft assignment, hard assignment can be done based on the highest probability.

Question 11/20

Which Python library is commonly used for geospatial data manipulation?

☐ scikit-learn

☐ matplotlib

☐ seaborn

☒ geopandas

Explanation:

Geopandas is specifically designed for handling geospatial data, allowing for efficient manipulation and visualization.

Question 12/20

True or False: Geopandas must be installed separately and is not included with Pandas by default.

☒ True

☐ False

Explanation:

Geopandas requires a separate installation as it is not bundled with Pandas.

Question 13/20

What type of data is stored in the 'geometry' column of a GeoDataFrame?

☒ Geographical data (points, lines, polygons)

☐ Time-series data

☐ Categorical data

☐ Numerical data

Explanation:

The 'geometry' column holds geographical information, including coordinates and shapes.

Question 14/20

Which method is used to read geospatial files into a GeoDataFrame in Geopandas?

☐ read_json

☒ read_file

☐ read_csv

☐ read_geojson

Explanation:

The read_file method in Geopandas is used to read various geospatial file formats into a GeoDataFrame.

Question 15/20

True or False: Hierarchical clustering can be used to group similar geographical regions based on demographic data.

☐ False

☒ True

Explanation:

Hierarchical clustering is a method that can be applied to group similar regions based on various data points, including demographic information.

Question 16/20

When visualising multi-dimensional data on a 2D plot, which technique can be used to represent an additional numerical feature?

☒ Varying the size of each point

☐ Altering the font type

☐ Changing the background colour

☐ Rotating the plot

Explanation:

- Varying the size of each point: Correct. Adjusting the size of points on a scatter plot is a common technique to represent additional numerical features.
- Changing the background colour: Incorrect. This does not effectively represent numerical features.
- Altering the font type: Incorrect. This is unrelated to visualising numerical features.

- Rotating the plot: Incorrect. This does not add any information about additional numerical features.

Question 17/20

What is the primary purpose of using the 'standard scaler' in preprocessing demographic data for clustering?

☐ To remove outliers from the dataset

☒ To normalise numerical data to have a mean of zero and a standard deviation of one

☐ To convert categorical data into numerical data

☐ To reduce the dimensionality of the data

Explanation:

The standard scaler is used to normalise numerical features, ensuring that they are on a similar scale.

Question 18/20

In hierarchical clustering, what does the 'ward' linkage criterion aim to minimise?

☐

Sum of squared distances from each point to the nearest cluster centroid

☐

Number of clusters



Total within-cluster variance

☐

Maximum distance between clusters

Explanation:

The 'ward' linkage criterion merges clusters based on minimising the total within-cluster variance, promoting more compact clusters.

Question 19/20

True or False: Geopandas can be used to overlay multiple layers of geographical data for comprehensive spatial analysis.



True

☐

False

Explanation:

Geopandas allows for the integration and analysis of multiple layers of geographical data, facilitating comprehensive spatial analysis.

Question 20/20

Which technique would you use to visualise the correlation between unemployment rates, average commute times, and income levels in different regions using a 2D plot?

☐

Use a pie chart



Plot a standard bar chart



A line chart



Create a 2D scatter plot with varying point sizes and colours

Explanation:

A 2D scatter plot with varying point sizes and colours allows for the visualisation of multiple dimensions on a 2D plot, effectively showing the correlation between different features such as unemployment rates, commute times, and income levels.